MD. Mehedi Hasan Maruf

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Summary

Software engineer with 3+ years of experience building Python backends (Django, FastAPI) and full-stack TypeScript systems. At Code Studio I shipped Django backends and ran the production Linux servers behind them. At Brain Station 23 I built StudentAI, a FastAPI and LangChain RAG learning platform, and at Aamira Software Solutions I build TypeScript microservices and a RAG feature. I also do applied ML research in Python: my B.Sc. thesis at Rajshahi University of Engineering & Technology (RUET) finds the university courses that hold students back and tests that answer with causal inference, and a paper from it is under review at LAK 2027.

Experience

Software Engineer II

Jan 2025 – Present

Aamira Software Solutions, Kokomo, Indiana, USA (remote)

- Lead three junior engineers on the company's React applications, after starting on its Angular project.
- Raised client response rates by 30% by leading development of a CRM system.
- Cut API response time by 40% by designing and building a TypeScript microservice backend.
- Made the application 60% faster by optimizing its Supabase (PostgreSQL) queries.
- Building a retrieval-augmented generation (RAG) feature for the platform.

Junior Software Engineer

Nov 2022 – Apr 2024

Code Studio, Rajshahi, Bangladesh

- Kept 12 Linux servers at 99.8% uptime, and cut deployment time by 40% by reworking the CI/CD pipelines.
- Moved 5 client environments onto Docker, which made them simpler to deploy.
- Wrote JavaScript, TypeScript, and Python (Django) backends used by thousands of people a day.

Industrial Trainee

Mar 2025

Brain Station 23, Dhaka, Bangladesh

- Represented a 15-person trainee cohort to management.
- Built and deployed StudentAI (see Projects), an AI learning assistant on FastAPI and Next.js.

Competitive Programming Mentor

Aug 2023 – Nov 2023

RAPL (University Organization)

- Taught competitive programming to RUET students, covering number theory and graph traversal algorithms.
- Organized biweekly coding contests.

Selected Projects

StudentAI | Intelligent Learning Companion

Mar 2025

- An AI learning assistant (FastAPI, Next.js) built during industrial training at Brain Station 23.
- Answers questions about uploaded documents through a LangChain RAG pipeline with semantic retrieval and conversation memory; it also analyzes YouTube lectures and writes quizzes.
- Accepts PDFs, images, and video, and has real-time WebSocket chat, JWT login, and progress tracking.
- Deployed on Vercel and Render with a Supabase database.

Thesis Site | Django Web App

Mar 2024 – May 2024

- Built a Django web app where students post thesis topics and talk with supervisors and the department.
- Built a responsive Bootstrap interface with full CRUD operations for students, supervisors, and administrators.

Skills

AI/ML: LLM applications, RAG, LangChain, Hugging Face Transformers, PyTorch, NLP, vector search (pgvector), model evaluation (RLHF)
Languages: Python, TypeScript, JavaScript, C++
Backend: FastAPI, Django, NestJS, Fastify, tRPC, .NET
Frontend: Next.js (React), Angular
Databases: PostgreSQL (Supabase), MongoDB, MySQL
DevOps: Docker, GitHub Actions (CI/CD), Linux server administration

Education

B.Sc. Engineering, Computer Science and Engineering *Mar 2022 – Aug 2026*

Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh

- CGPA: 3.75 / 4.00, Honours
- Thesis: *TRACE*, which ranks the courses that hold students back using prerequisite graphs, Bayesian fail-rate models, and student-feedback sentiment, then checks the result with causal inference on 23,369 student records. A follow-up paper is under review at LAK 2027.

Certifications & Achievements

- Digital Accessibility Training (DAT) — a2i (Aspire to Innovate), Govt. of Bangladesh (Oct 2025)
- Industrial Training Certificate — Brain Station 23 (Mar 2025)
- PHP (Laravel) — EDGE Project (Feb 2025)
- Experience Certificate — Code Studio (Apr 2024)
- 4th Place — RUET CodeSmash, team competitive programming contest (Mar 2023)
- 4th Place — RUET CodeSpark, individual competitive programming contest (Oct 2022)